

BookSmart

Predicting Optimal Booking Timing for Intercity Bus Travel Using Web-Scraped Fare Data

Student name	Deepak S
Register number	CB.SC.U4CSE23267
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Repository contents	README.md, data/ (raw and cleaned CSV), analysis.ipynb, Case_Study_Report.pdf

1. Problem Statement and Objectives

Intercity bus operators in India price seats dynamically. The fare quoted for the same route, operator, bus type and departure time changes depending on when the ticket is bought. Individual travellers, corporate travel desks and institutions that book buses in bulk usually decide when to buy on instinct. Some book weeks in advance on the assumption that fares only rise, while others wait for a last-minute deal. Neither habit is based on evidence for the route and operator in question, so travellers either overpay or miss cheaper windows.

The problem matters because intercity buses are the main long-distance mode for a large share of Indian travellers, online bus ticketing is a large and growing market, and a corporate travel desk books hundreds of such trips a month. A few percent saved per ticket adds up across a travel budget. It also matters to ticketing platforms, which can build trust by giving buyers clear timing advice, and to operators, who can compare their pricing with rivals on the same corridor.

Objectives

1. Identify the factors that drive quoted intercity bus fares: days to departure, seat occupancy, operator, bus class, departure slot, travel day and festival periods.
2. Build a regression model that predicts the expected fare of a bus listing.
3. Build a classification model that labels a quoted fare Book Now (clearly cheaper than what the same bus charges close to departure) or Wait.
4. Measure how the booking-window effect differs across operators and corridors, and turn the results into recommendations.

2. Data Collection and Dataset Description

Method and source. The data was collected by web scraping publicly accessible pages of redBus (www.redbus.in), India's largest online bus-ticketing platform. A route search page such as redbus.in/bus-tickets/bangalore-to-chennai lists every bus on sale for a chosen date and can be viewed without an account. No ready-made dataset was used.

Collection procedure. Eight corridors were chosen to cover South, West and North India and a mix of short and overnight trips: Bengaluru-Chennai, Bengaluru-Hyderabad, Bengaluru-Coimbatore, Chennai-Coimbatore, Chennai-Madurai, Hyderabad-Vijayawada, Mumbai-Pune and Delhi-Jaipur. For each corridor, listings were collected for 13 departure dates at 1, 2, 3, 4, 5, 6, 7, 10, 14, 21, 30, 45 and 60 days after the scrape date, which gives 104 route-date searches. The search page fills its results from a JSON endpoint that the site's own front end calls. The same endpoint was queried with paging of 100 results, so every listed bus was captured, not just the first screen. Government operators (APSRTC, TGSRTC, KSRTC, RSRTC and others) appear as collapsed groups on the page, so a second pass expanded each group. Script requests (curl or Python) could not reach the site from the analysis environment, so the collector ran inside a normal browser

session on the site, with a 1.2-1.5 second pause between requests. The collector code is kept in the notebook for reproducibility.

Size and attributes. The raw file in data/raw/ holds 32,372 listings with 19 attributes from 981 operators. One record is one bus service on sale for one route and departure date. After cleaning, 31,838 records remain (data/cleaned/bus_fares_clean.csv, 27 columns including derived features).

Table 1. Key variables

Variable	Type	Description
fare (fare_min)	Numeric (Rs)	Lowest seat fare shown for the listing - the price a buyer sees first. Target variable.
days_to_departure	Numeric	Departure date minus scrape date (1-60). The booking window.
seats_available, total_seats, occupancy	Numeric	Seats left on redBus, bus capacity, and share of the redBus seat map already sold (1 - available/total).
operator, is_govt	Categorical / flag	Operator name; government transport corporation vs private operator.
bus_type, bus_class, is_ac, is_sleeper	Categorical / flags	Raw bus description and a derived class (AC/Non-AC x Seater, Sleeper, Seater/Sleeper).
departure_time, dep_slot, duration_hrs	Time / numeric	Departure time, time-of-day slot, journey length.
rating, num_reviews	Numeric	Passenger rating (1-5) and number of reviews shown on the listing.
day_of_week, is_weekend_dep, is_festival	Categorical / flags	Travel day, Friday-Sunday flag, and festival-period departure flag.
route, service code	Categorical / ID	Corridor, and the operator's schedule code used to follow the same bus across dates.

Table 2. Records per corridor after cleaning

Corridor	Listings	Operators	Median fare (Rs)	Mean occupancy
Hyderabad - Vijayawada	6,817	166	649	17%
Delhi - Jaipur	2,856	93	700	16%
Bengaluru - Chennai	2,672	73	949	23%
Mumbai - Pune	6,712	317	1,019	12%
Bengaluru - Coimbatore	2,704	136	1,200	26%
Chennai - Coimbatore	2,242	115	1,253	30%
Chennai - Madurai	4,514	172	1,400	29%
Bengaluru - Hyderabad	3,321	109	1,592	20%
Total	31,838	981	1,056	20%

Personal data. Only public listing information was collected. There was no login, no booking and no passenger data. The service code identifies a bus schedule, not a person, and it is replaced by an integer key in the cleaned file.

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and preprocessing

- **Type conversion:** fares, seats, ratings and review counts cast to numbers; journey minutes converted to hours.
- **Invalid rows:** 9 listings with no bookable fare or zero capacity removed.
- **Duplicates:** a service was identified by operator + operator service code + departure time + corridor; 6 repeated rows removed. The operator is part of the key because some operators use short numeric codes (for example "241") that clash with other operators.

- **Missing values:** 992 listings with no rating yet were given the operator's median rating (global median as fallback); missing review counts set to 0.
- **Derived features:** days to departure, travel day, weekend flag, departure slot, occupancy, bus class, government flag, and a festival flag for departures in the Gandhi Jayanti long weekend (1-4 Oct), the Dussehra week (16-21 Oct) and the Diwali-Chhath period (5-16 Nov 2026). 30% of listings fall on festival dates.
- **Outliers:** fares outside the 1st-99th percentile within each corridor removed (519 rows), which drops glitches such as placeholder fares while keeping genuine premium buses.
- **Rare categories:** operators with fewer than 40 listings grouped as "Other operators" for modelling (152 groups).

3.2 Descriptive analysis

The median starting fare is Rs 1,056 and the mean is Rs 1,382. The distribution is right-skewed (skewness 1.91), so the regression models use log fare (skewness 0.21). Bus class separates fares most clearly: the median Non-AC seater costs Rs 440 and the median AC sleeper Rs 1,399. Longer overnight corridors (Bengaluru-Hyderabad, Chennai-Madurai) are the most expensive. Government buses are much cheaper than private ones (median Rs 533 vs Rs 1,199), mostly because they run more seater and non-AC buses.

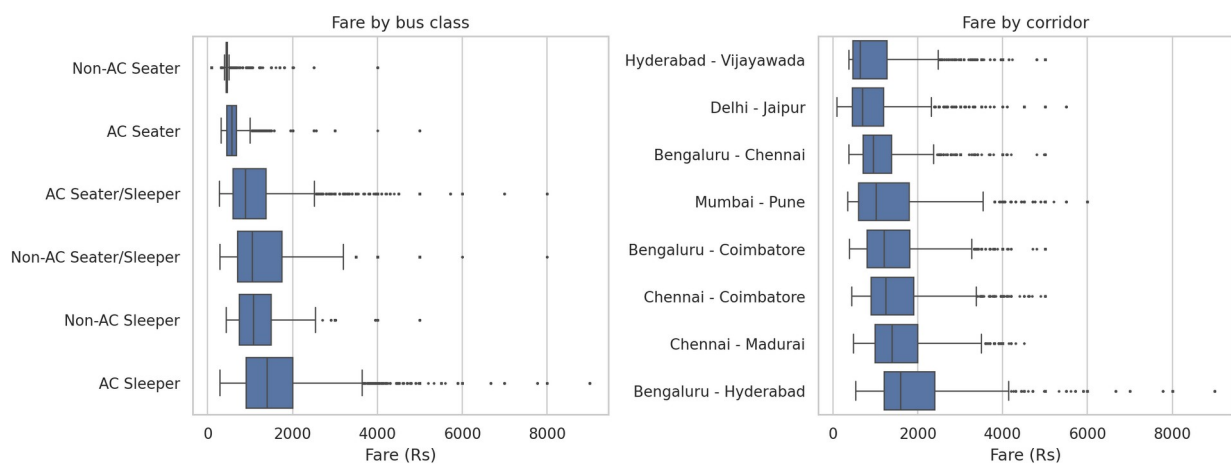


Figure 1. Starting fare by bus class and by corridor (all dates).

Figure 2 shows Spearman correlations between fare and the numeric features. Sleeper buses (0.46) and government operation (-0.34) have the strongest links with fare, followed by days to departure (0.27), festival departure (0.24) and journey length (0.20). Occupancy is almost uncorrelated with fare (0.02) even though it rises as departure approaches (-0.21 with days to departure), which is a first sign that operators here do not raise fares as buses fill. Days to departure and the festival flag are correlated (0.39) because three of the thirteen sampled dates fall in festival periods, so the two effects have to be separated. The next section does this.

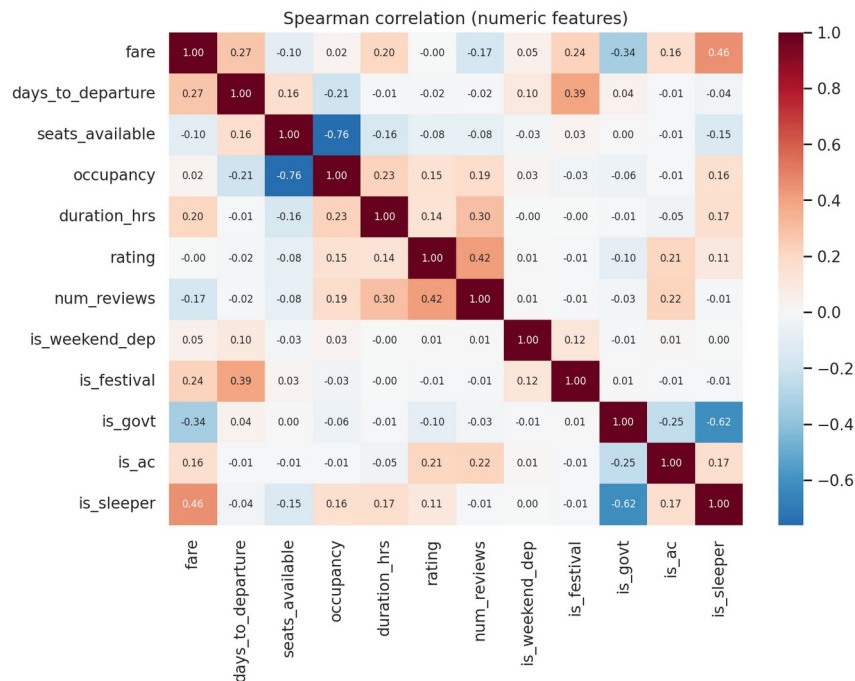


Figure 2. Spearman correlation between fare and numeric features.

3.3 The key relationship: fare versus booking window

A simple average of fares by departure date is misleading, because the set of buses changes: 60 days out only about 500 buses are on sale, while 1 day out there are about 2,800. To isolate timing, each listing's fare was divided by the median fare of the same bus service across all 13 dates (index 100 = that bus's usual fare). This holds route, operator, bus type and departure time fixed. 3,073 services were observed on at least five dates (30,391 listings). Medians are used because a few services post very high placeholder fares far in advance, which pull averages up.

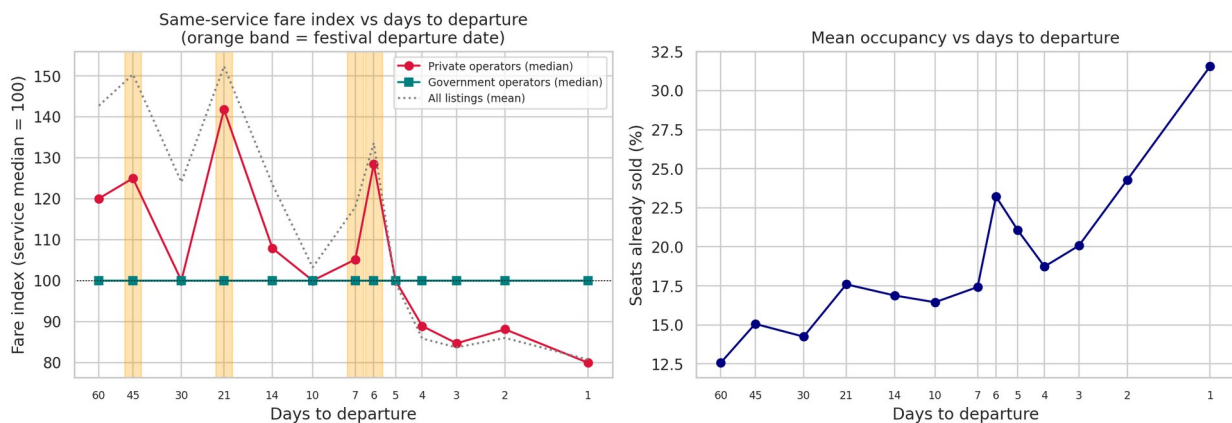


Figure 3. Same-service fare index by days to departure (left) and share of redBus seats already sold (right). Orange bands mark festival departure dates.

The pattern is the opposite of the common belief that fares rise as departure nears. For private operators on ordinary dates, the median index stays at 100 from 60 days out to 5 days out, then falls to 80-89 in the final four days. On ordinary dates, private fares are 14.3% lower in the last four days than earlier. The seat maps point to the reason: the day before departure only 31% of the seats shown on redBus are sold on average (12% at 60 days out), so operators discount the seats still unsold online. This figure counts redBus inventory only; operators also sell through counters, agents and their own apps, and redBus reports 77% average final occupancy for October 2025 to March 2026 [10], so buses do not leave this empty. Government operators do not move at all (index 100 on every date). Festival departures are the exception: the median private fare on festival dates is 20% above normal, and up to 42% above normal for the Friday before Dussehra week.

Near and far dates fall on different weekdays, so the comparison was repeated for pairs of dates with the same weekday. The drop remains (Table 3), which shows the effect comes from timing rather than the day of the week.

Table 3. Same private bus services, same weekday, near vs far booking window

Travel day	Near (days out)	Far (days out)	Services in both	Median change near vs far	Cheaper near / dearer near
Monday	3	10	2,287	-9.1%	56% / 8%
Sunday	2	30	1,110	-25.0%	70% / 7%
Tuesday	4	60	300	-48.6%	72% / 4%

The Tuesday gap is inflated by the high advance fares some operators post 60 days out, so the Monday pair (3 vs 10 days) is the more conservative estimate. Two other patterns show up: Friday departures carry a premium (the non-festival Friday, 9 October, sits about 8% above normal in the median), and afternoon and evening departures cost more on average than morning ones.

4. Analytics Method and Implementation

Why these methods. The study has one continuous question (what should this listing cost?) and one decision question (buy now or wait?). Both are supervised learning problems from the Business Analytics syllabus. Multiple linear regression gives an interpretable baseline whose coefficients read as percentage effects on log fare. Tree ensembles (Random Forest and Gradient Boosting) capture the non-linear booking curve and interactions such as bus class by corridor without manual feature design, and they report feature importance, which answers the business question of what drives fares. For the decision, logistic regression is the interpretable baseline and a Random Forest classifier is the flexible model.

Implementation. All work is in analysis.ipynb (Python, pandas, scikit-learn [8]). Features: days to departure, occupancy, seats available, duration, rating, number of reviews, weekend, festival and government flags (numeric), plus corridor, operator group, bus class, departure slot and travel day (one-hot encoded). The regression target is log fare. Data is split 80/20 at random with a fixed seed, and 5-fold cross-validation on the training part checks stability. Random Forest uses 300 trees [9]; Gradient Boosting uses scikit-learn's histogram-based booster (400 iterations, learning rate 0.06).

Book Now / Wait label. The label is built from observed fares, not from a model. For every service seen 1-3 days before departure, its near-departure fare is the mean of those observations. A listing of the same service 4 or more days out is labelled Book Now if its fare is at least 5% below that near-departure fare, and Wait otherwise. The 5% band ignores trivial differences. Festival departures are excluded, because the 1-3-day reference dates are all ordinary days and would not represent a real alternative. This leaves 13,010 labelled listings from 3,040 services, 11.6% of them Book Now. Because the classes are imbalanced, the classifiers use balanced class weights and a stratified split, and they are judged by F1 for the Book Now class and ROC-AUC, against a majority-class baseline.

5. Comparison with State-of-the-Art Methods

Five published studies on dynamic transport pricing were reviewed, three of them from 2021 onward. Table 4 follows the prescribed format.

Table 4. State-of-the-art comparison

Published study / Year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
Gaggero, Ogrzewalla & Bubalo [2], 2019	Fares scraped daily from the Flixbus website, 5 city pairs (4 German, 1 Germany-Italy), 28 to 1 days before departure	Panel fixed-effects regression with endogeneity and sample-selection corrections	Coefficient estimates and significance	Fare rises with seats sold; the lowest available fare rises as departure nears; no evidence of intertemporal price discrimination	Closest design to this study (scraped posted fares across the booking window). Their single-operator market shows rising fares; this multi-operator Indian market shows last-minute cuts on ordinary dates. This study adds prediction and a buy/wait classifier; theirs gives stronger causal controls.
Branda, Marozzo & Talia [3], 2020	3.23 million event-log records from a bus ticketing platform	Process mining; Naive Bayes, Logistic Regression, Decision Tree, Random Forest, XGBoost; simulated pricing strategy	Accuracy; ticket and revenue uplift	95% accuracy predicting ticket purchase; pricing strategy +6% tickets and +9% revenue	Seller-side view using private logs; this study is buyer-side using only public listings. The targets differ (purchase vs fare/timing), so the 95% accuracy cannot be compared with this study's scores.
Stavinova, Chunaev & Bochenina [4], 2021	Renfe (Spain) crawled rail fares, Madrid-Barcelona, 75 daily observations (Apr-Jun 2019)	ARIMA, LSTM, ARIMAX, multivariate LSTM with Google Trends queries	RMSE, MAPE	Multivariate LSTM with search data: MAPE 3.67% vs 4.25% baseline for mean daily price, 4 days ahead	Forecasts one aggregate daily series; this study models 31,838 individual listings across 8 routes. Their external demand signal (search trends) is a useful addition for festival demand here.
Degife & Lin [5], 2023	1,032,215 Ethiopian Airlines records, 22 destinations, 2018-2022, 44 features incl. season, weekend, holiday	GRU compared with LSTM, MLP and classic ML	MAE, RMSE, R2	GRU outperformed LSTM, MLP and classic models on all three metrics	Carrier-internal data at far larger scale, with deep learning. This study uses public data and interpretable tree models (R2 0.83 on log fare). Both find holiday and weekend effects; scores are not comparable across datasets.
Arneric & Obadic [6], 2026	FlixBus fares on 24 Croatian routes (16 international, 8 domestic), monthly, 2022	Panel fixed, random and two-way effects models with poolability tests	Model diagnostics and elasticities	Distance elasticity below 1; advance-booking price discrimination confirmed; seat occupancy not significant	Also finds that occupancy does not drive posted fares directly. Here, the linear model even gives occupancy a negative sign, consistent with operators discounting unsold seats rather than pricing on load.

Strengths and limitations relative to published work. This study's strengths are a fresh, public, multi-operator dataset from the Indian market, a same-service design that separates timing from bus mix, and a direct buyer-facing decision model. Its main limitation against the panel studies [2], [6] is that it uses a single scrape day, so it compares different departure dates rather than following one departure's price over time. As the brief notes, a higher score in another paper does not mean a better method, because the datasets, targets and evaluation settings differ.

6. Results, Business Insights and Recommendations

6.1 Fare regression

Table 5. Regression results (test set, 6,368 listings)

Model	CV R2 (log)	Test R2 (log)	MAE (Rs)	RMSE (Rs)	MAPE
Linear Regression	0.575	0.575	475	786	34.6%
Random Forest	0.801	0.830	282	517	18.8%
Gradient Boosting	0.780	0.789	331	579	22.5%

The Random Forest explains 83% of the variation in log fare, with a typical error of about Rs 280 (19%). Cross-validation (0.80) is close to the test score, so the model is stable. The linear model is much weaker (R2 0.58), which confirms the relationship is non-linear. Bus class, operator, days to departure and corridor are the strongest drivers. In the linear model, a government operator is 23% cheaper for an otherwise similar listing,

and each extra day of lead time adds about 1.1% to the fare, which matches the last-minute cuts seen in Section 3. The model underpredicts some very expensive listings (above Rs 4,000), which are mostly AC sleepers quoted five or more days ahead, many of them on festival dates.

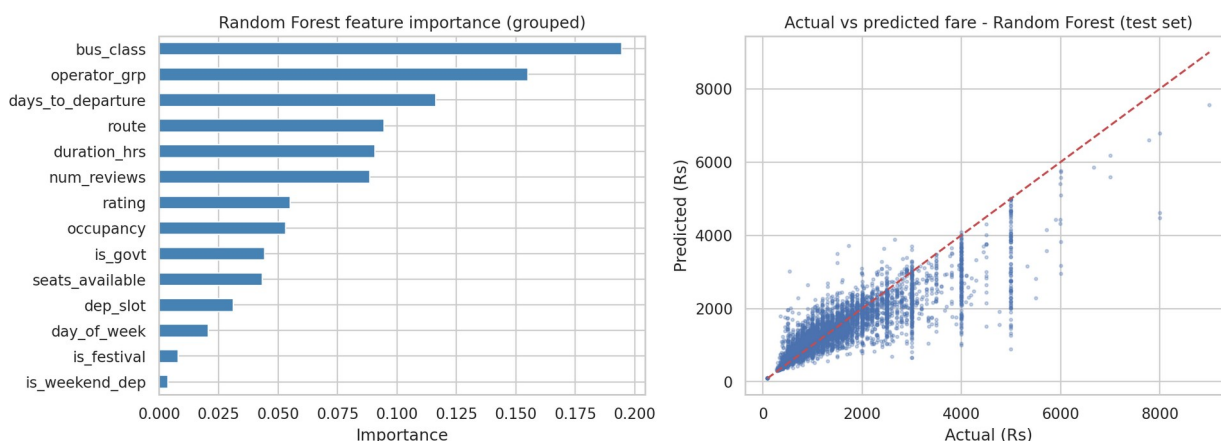


Figure 4. Random Forest feature importance (grouped) and actual vs predicted fare on the test set.

6.2 Book Now / Wait classifier

Table 6. Classification results (test set, 2,602 listings, 302 Book Now)

Model	Accuracy	F1 (Book Now)	ROC-AUC
Majority-class baseline (always Wait)	0.884	-	0.500
Logistic Regression	0.733	0.407	0.844
Random Forest	0.864	0.546	0.893

Always saying Wait is 88% accurate but never finds a bargain, so accuracy alone is misleading here. The Random Forest ranks listings well (ROC-AUC 0.89) and catches 71% of the true Book Now cases, with 45% precision. For a fare-guidance badge this is a reasonable trade-off: a false Book Now costs the buyer little, because Wait listings are typically only about 14% (Rs 105) dearer than the last-minute fare, while a missed bargain loses a median saving of 14.3% (Rs 120).

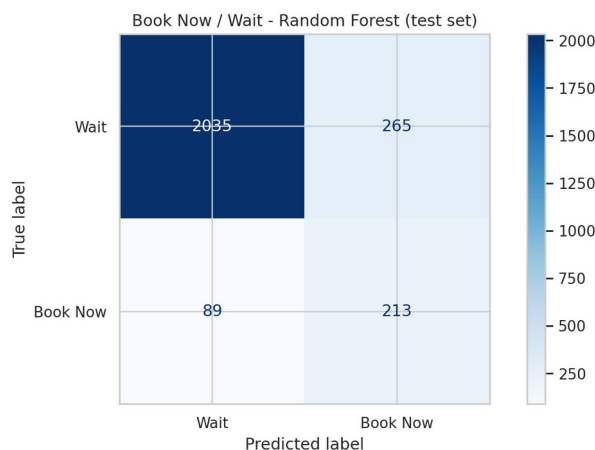


Figure 5. Confusion matrix of the Random Forest classifier (test set).

6.3 Business insights

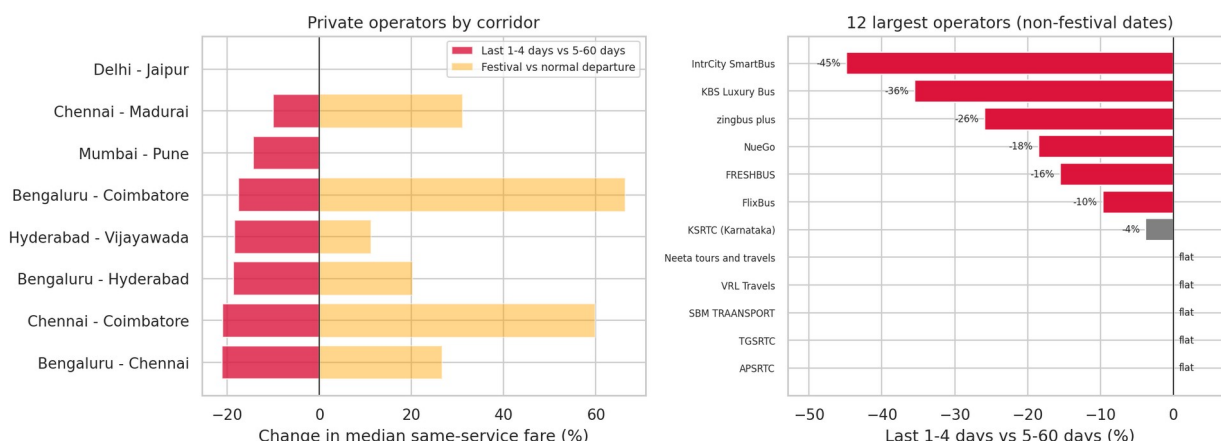


Figure 6. Left: last-minute change and festival premium by corridor (private operators). Right: last-minute change for the 12 largest operators.

- **Late booking is usually cheaper, not dearer.** Of the 56 operators with enough data, 36 cut fares by more than 5% in the last four days, 20 stay flat and none raise them. The deepest cuts come from large app-first private brands such as IntrCity SmartBus (-45%), KBS Luxury Bus (-36%), zingbus plus (-26%), NueGo (-18%), FRESHBUS (-16%) and FlixBus (-10%). Older private operators such as VRL Travels and SBM Transport keep fares flat, as do the state corporations.

Table 7. Twelve largest operators - median same-service fare index, ordinary dates

Operator	Type	Listings	Index 5-60 days out	Index 1-4 days out	Last-minute change
APSRTC	Government	1,112	100.0	100.0	0%
TGSRTC	Government	1,012	100.0	100.0	0%
FlixBus	Private	762	100.0	90.3	-9.7%
IntrCity SmartBus	Private	630	100.3	55.2	-44.9%
VRL Travels	Private	372	100.0	100.0	0%
NueGo	Private	330	103.5	84.4	-18.5%
SBM Transport	Private	297	100.0	100.0	0%
KBS Luxury Bus	Private	295	100.0	64.5	-35.5%
KSRTC (Karnataka)	Government	286	100.0	96.2	-3.8%
zingbus plus	Private	251	104.6	77.5	-25.9%
Neeta Tours and Travels	Private	243	100.0	100.0	0%
FRESHBUS	Private	221	100.0	84.5	-15.5%

- **Festivals reverse the rule.** Private fares on festival departures are 20% above normal in the median, and 60-66% above normal on the Coimbatore corridors. Last-minute discounting cannot be relied on for these dates.
- **Corridors differ.** The last-minute drop is largest on Bengaluru-Chennai and Chennai-Coimbatore (-21%) and smallest on Chennai-Madurai (-10%). Delhi-Jaipur shows no timing effect. Bengaluru-Chennai has the most genuine Book Now chances (25.5% of listings); Hyderabad-Vijayawada, dominated by state buses, has almost none (2.5%).
- **Timing hardly matters for government buses.** Their fares are fixed (Book Now in only 1.2% of cases), so buyers can book them purely for convenience.

6.4 Recommendations

- **Travellers and corporate travel desks:** on ordinary dates, do not book private AC sleepers weeks ahead. Book 1-4 days before travel, when fares are about 14% lower and most online seats are still

unsold. Book festival and Friday departures early, when advance fares are the safer choice. Book government buses whenever convenient. A travel desk can apply these rules as policy and use the classifier to flag the exceptions.

- **Ticketing platforms:** show a fare-guidance badge driven by the Book Now / Wait model. It should say "likely cheaper closer to the date" as often as "good price now", which is more honest than urgency messages, and pair it with a festival calendar.
- **Operators:** deep last-minute cuts show that advance pricing on ordinary dates is set too high for demand (only 12-18% of online seats are sold two weeks or more before departure). Lower early fares or early-bird tiers would fill seats sooner. The discounts also teach regular travellers to wait, so operators with flat pricing should track whether they lose late bookers to rivals that discount.

7. Conclusion and References

Using 31,838 real redBus listings across eight corridors and a 60-day booking horizon, this study finds that intercity bus fares in India do not follow the airline-style pattern of rising prices as departure nears. For private operators on ordinary dates, the same bus is about 14% cheaper in the last four days, because operators discount seats still unsold online close to departure. Government fares are fixed. Festival departures carry a 20% or larger premium, and there the usual advice to book early does apply. A Random Forest predicts fares with R^2 0.83 on log fare, and a Book Now / Wait classifier with ROC-AUC 0.89 turns these patterns into a usable buying signal. The objectives were met, and the main finding contradicts the intuition that motivated the study, which is the kind of correction evidence-based booking is meant to provide.

Limitations and future work. (1) All data comes from one scrape day, so the analysis compares different departure dates rather than following one departure's price over time; repeated daily scrapes would give a true longitudinal panel. (2) Near and far departure dates differ in season, which the same-weekday comparisons reduce but cannot remove. (3) The starting fare is the cheapest seat, not the average seat. Occupancy is measured from the redBus seat map only, not the operator's full sales. (4) Festival windows are defined by calendar, not measured demand; search-trend data as in [4] could improve this.

References

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